SILTHAUL · MARKET AND VIABILITY

SiltHaul market and viability

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First market note, written for the kit 1.9.1 upgrade; prices cited with the date they were checked

SiltHaul market and viability

Relief groups and municipal crews that clear flood mud by shovel and bucket are the users; no portable, hand-powered slusher for flood recovery was found (SLH-PRB-001), so the benchmarks below are partial and only show the price range of hand pulling and hand pumping gear. An open design lets a local fabricator build and mend a kit from stock steel for about USD 1,056 in parts, and spread it through NGOs and workshops without a licence fee. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Relief NGOs and community groups	Flood clean-up in South and Southeast Asia, for example Kerala after the 2018 floods (Gulf News, 2018), and other flood regions	Shovels and buckets (no price checked: to assess); a 2 ton hand cable puller (come-along), USD 83.99, listed as discontinued, on a retailer page checked 2026-11-03 (Paul B Hardware); its 6 ft cable cannot drag a box 12 m	Option A: 1,055.60 (BOM, indicative prices). Option B: about 454 for lines carried over from the BOM, foot boards and grips not priced. Option C: to assess	A kit that two or three volunteers can set up and a crew of four can use to move mud without lifting it
Municipal drain and sanitation crews	Narrow lanes, open drains and culverts where machines cannot enter	Powered mine slushers and scraper winches (Mining History Association); no price found: to assess	As above	A hand-powered rope scraper with no fuel or power supply
Householders and volunteers	Owners of flooded ground-floor rooms	No sale: users borrow a kit from an NGO or council	Not applicable	Less lifting and less time in the mud
Local fabricators	Workshops near flood-prone areas	A hand diaphragm pump, for scale only: AMT hand diaphragm pump HP01-99, USD 182.95, USA Blue Book page checked 2026-11-03 (USA Blue Book); the page does not say whether it passes solids, and it is far smaller than option C	Parts only; labour and workshop margin to assess	A paid build and repair service from open drawings

Open route

NGOs and relief groups would borrow or buy a kit, local fabricators would build it from `bom/bom.csv` and the drawings, and universities and makerspaces could build a test unit. The hardware is under CERN-OHL-S-2.0 and the calculators under MIT (`REUSE.toml`), so any maker can share improvements back under the same terms. Whether a commercial supplier would ever sell kits is Amish's decision and is not assumed here.

Viability

On paper the parts cost is USD 944.40 under the USD 2,000 value-engineering target (a hypothetical control target). Demand is episodic: flood seasons, and one kit serves many sites if the NGO stores it. Unknown: willingness to pay, how many kits an NGO would hold, and local labour rates; all are "to assess" with a co-design partner. No price in this note is a quote.

Sources

- Gulf News, 2018, Kerala clean-up, https://gulfnews.com/world/asia/india/kerala-flood-70000-people-participate-in-massive-post-flood-clean-up-drive-in-kuttanad-1.2271770.
- Paul B Hardware, Maasdam Powr-Pull 2 ton cable puller MP144SB6, USD 83.99, discontinued, checked 2026-11-03.
- USA Blue Book, AMT hand diaphragm pump HP01-99, USD 182.95, checked 2026-11-03.
- Mining History Association, journal 2015 on slusher scrapers, https://www.mininghistoryassociation.org/Journal/MHJ-v22-2015-Reynolds.pdf.